STAT/MA 41600
In-Class Problem Set #32 Part 2: November 2, 2026
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Problem Set 32 Part 2 Answers

1a. Since exponential random variables have the memoryless property, we know that $P(X_3 > 2.5 \mid X_3 > 0.75) = P(X_3 > 2.5 - 0.75) = P(X_3 > 1.75) = e^{-(2)(1.75)/(3)} = 0.3114$.

Alternatively, we can compute $P(X_3 > 2.5 \mid X_3 > 0.75) = \frac{P(X_3>2.5 \ \& \ X_3>0.75)}{P(X_3>0.75)} = \frac{P(X_3>2.5)}{P(X_3>0.75)} = \frac{e^{-(2)(2.5)/(3)}}{e^{-(2)(0.75)/(3)}} = e^{-(2)(1.75)/(3)} = 0.3114$.

1b. For $a > 0$, we have

$$\begin{aligned} P(Y > a) &= P(\min(X_1, X_2, X_3) > a) \\ &= P(X_1 > a \ \& \ X_2 > a \ \& \ X_3 > a) \\ &= (e^{-2a/3})(e^{-2a/3})(e^{-2a/3}) \\ &= e^{-2a}. \end{aligned}$$

It follows that $F_Y(a) = P(Y \leq a) = 1 - e^{-2a}$ for $a > 0$, and otherwise $F_Y(a) = P(Y \leq a) = 0$.

So the probability density function of Y is $f_Y(y) = 2e^{-2y}$ for $y > 0$, and $f_Y(y) = 0$ otherwise.

2a. The number of Vice Presidents who repost the article within 1 hour is a Binomial random variable with $n = 3$ and $p = P(X_i < 1) = 1 - e^{-2/3} = 0.4866$.

2b. Let X denote the number of Vice Presidents who repost the article within 1 hour. Then the probability mass function of X is $p_X(x) = \binom{3}{x}(1 - e^{-2/3})^x(e^{-2/3})^{3-x}$. We have:

$p_X(0) = (e^{-2/3})^3 = e^{-2} = 0.1353$; $p_X(1) = 3(1 - e^{-2/3})^1(e^{-2/3})^2 = 0.3848$; $p_X(2) = 3(1 - e^{-2/3})^2(e^{-2/3})^1 = 0.3647$; $p_X(3) = (1 - e^{-2/3})^3 = 0.1152$; and these add to 1.

3a. For $a > 0$, we have $P(X_1 + X_2 \leq a) = \int_0^a \int_0^{a-x_1} (2/3)e^{-2x_1/3}(2/3)e^{-2x_2/3} \, dx_2 \, dx_1 = 1 - e^{-2a/3} - (2/3)(a)e^{-2a/3}$, and otherwise, $P(X_1 + X_2 \leq a) = 0$ for $a \leq 0$.

3b. For $a > 0$, we have

$$\begin{aligned} P(X_1 + X_2 + X_3 \leq a) &= \int_0^a \int_0^{a-x_1} \int_0^{a-x_1-x_2} (2/3)e^{-2x_1/3}(2/3)e^{-2x_2/3}(2/3)e^{-2x_3/3} \, dx_3 \, dx_2 \, dx_1 \\ &= 1 - e^{-2a/3} - (2/3)(a)e^{-2a/3} - (2/9)(a^2)e^{-2a/3}, \end{aligned}$$

and otherwise, $P(X_1 + X_2 + X_3 \leq a) = 0$ for $a \leq 0$.

4a. For $a > 0$, we have $P(\max(X_1, X_2) \leq a) = (1 - e^{-2a/3})^2$, and $P(\max(X_1, X_2) \leq a) = 0$ for $a \leq 0$.

4b. For $a > 0$, we have $P(\max(X_1, X_2, X_3) \leq a) = (1 - e^{-2a/3})^3$, and $P(\max(X_1, X_2, X_3) \leq a) = 0$ for $a \leq 0$.

5a. We have $0 < U < 1$, and therefore $-\infty < \ln U < 0$, so we conclude $0 < -\frac{1}{7}\ln(U) < \infty$. So X takes on positive real values.

5b. For $a \leq 0$, we have $F_X(a) = 0$. For $a > 0$, we compute $F_X(a) = P(X \leq a) = P(-\frac{1}{7}\ln(U) \leq a) = P(\ln(U) \geq -7a) = P(U \geq e^{-7a}) = 1 - e^{-7a}$.

5c. Using the CDF of X from **5b**, we see that X is an Exponential random variable with $\lambda = 7$, i.e., with average $1/7$.